

1. Literature Review

Legal recommender systems go back further than one might expect. Winkels et al. [1] built one of the earlier examples for a Dutch legislative portal, where the system pointed users toward related statutes by turning the references already present in a case into machine-readable links. Their approach

worked well for cases that already cited relevant law, but it relied on citation structure rather than on what the case was actually about, so two factually similar cases with no shared citations would not be linked at all.

A more practitioner-oriented system is Quick Check, described by Thomas et al. [3]. It reads a litigation brief, extracts the legal arguments from it, and recommends case law opinions using a mix of full-text search, citation-network analysis, and clickstream data, with a ranking model trained on over ten thousand human annotations. It is a strong system, but it needs a large annotated corpus and an established citation graph behind it, resources that are simply not available for a small, manually built dataset of the kind used in this project.

Closer to the approach taken here, Dhanani, Mehta and Rana [2] built a legal document recommender for Indian court judgments using a dictionary-based representation instead of the heavier Doc2Vec embeddings used in earlier systems, arguing that the sheer size and noise of a legal vocabulary makes embedding models slow to train without necessarily improving the results. Makawana and Mehta [9] took the representation question further and proposed a paragraph-level filtering technique that treats a judgment as a network of paragraphs rather than one flat block of text, using that structure to decide which parts of two documents are worth comparing at all.

Other recent work moves past plain word-overlap methods altogether. Dhani et al. [4] built a knowledge graph out of Indian court judgments and framed case similarity as a graph problem, linking cases through shared statutes, sections and cited precedents rather than shared vocabulary. Oliveira and Sperandio Nascimento [6] fine-tuned several transformer models, BERT, GPT-2, RoBERTa and LLaMA, on over two hundred thousand Brazilian labour-court cases and clustered the resulting embeddings; the domain-tuned LLaMA model produced noticeably tighter, more accurate clusters than any of the classic NLP baselines they compared it against. Ostendorff et al. [7] ran a broader comparison still, testing twenty-seven different document representation methods, text-based, citation-based and hybrid, on a benchmark of US case law they built themselves, and found that no single representation dominates across the board, which is part of why the present project sticks with a simple, explainable TF-IDF baseline rather than reaching straight for a transformer model. Li et al. [8] approached the retrieval side of the problem differently again, releasing LeCaRDv2, a large Chinese legal case retrieval benchmark with over fifty thousand candidate cases judged on characterization, penalty and procedure rather than plain topical overlap, which is a useful reminder that matching subject matter alone does not capture what makes two cases relevant to each other.

Large language models are the newest addition to this line of work. Shu et al. [5] introduced LawLLM, a model built specifically for the US legal system that handles similar-case retrieval, precedent recommendation and judgment prediction as three separate tasks, on the grounds that researchers often blur the line between a similar case and a precedent case when really the two call for different techniques. Deroy et al. [10] look at the problem from a data-organisation angle instead, discussing how legal ontologies, citation graphs and TF-IDF based topic extraction can be combined to make an otherwise unstructured mass of judgments usable by both practitioners and downstream systems.

Taken together, this body of work shows a fairly clear split: citation-graph and knowledge-graph methods work well once a case corpus has rich cross-references, while transformer and LLM-based methods need either a large labelled dataset or heavy compute, or both. Neither condition holds for a small, freshly curated set of Hindu Marriage Act judgments with no existing citation graph. A TF-IDF and cosine-similarity baseline, of the kind used by Dhanani et al. [2] in the Indian legal setting, is therefore a reasonable starting point: it needs no training data beyond the corpus itself, it stays fully explainable, and it can be built and tested quickly enough to fit within a course project's timeline. That is the gap this paper's system sits in, and Section 3 describes how it was built.